

GADSDEN 19 N, AL, USA

WMO: 722380

Lat: 34.2851N

Lon: 85.9621W

Elev: 351

StdP: 97.18

Time Zone: -6.00 (NAC)

Period: 05-19

WBAN: 63857

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-8.2	-5.8	-14.4	1.1	-6.2	-12.1	1.4	-4.2	7.2	3.1	6.6	2.8	1.8	N/A	0.346

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.2	33.1	22.9	31.9	22.9	30.8	22.8	24.9	29.9	24.4	29.4	23.9	28.8	2.1	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
23.5	19.1	27.3	22.9	18.5	26.9	22.5	17.9	26.3	77.5	30.0	75.5	29.4	73.5	28.8	27.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 6.0	5.3	4.6	-11.6	34.9	2.7	1.8	-13.6	36.2	-15.2	37.3	-16.7	38.4	-18.7	39.7		
(5)			-12.5	25.9	3.0	0.7	-14.6	26.4	-16.4	26.8	-18.1	27.2	-20.3	27.7	(4)	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	15.3	4.1	6.3	10.8	15.1	19.3	23.6	24.8	24.7	21.9	15.6	9.5	6.8
	DBStd	8.41	5.72	5.66	5.07	4.06	3.54	2.06	1.96	2.15	3.02	4.59	4.66	5.28
	HDD10.0	589	197	130	53	8	0	0	0	0	0	8	64	130
	HDD18.3	1897	441	337	236	109	33	1	0	0	8	107	266	358
	CDD10.0	2508	14	27	77	161	288	407	458	457	357	183	49	31
	CDD18.3	774	0	1	2	13	63	158	199	198	115	24	1	1
	CDH23.3	6317	0	1	16	86	486	1389	1630	1642	877	183	7	0
	CDH26.7	2201	0	0	1	3	92	507	618	656	288	36	1	0
Wind	WSAvg	1.8	2.3	2.5	2.4	2.0	1.3	1.2	1.3	1.3	1.4	1.7	2.0	2.3
Precipitation	PrecAvg	1416	142	130	150	112	120	100	110	95	107	79	114	134
	PrecMax	1669	255	292	264	193	339	230	202	204	218	212	271	235
	PrecMin	938	60	58	46	61	19	27	44	27	10	0	54	65
	PrecStd	206	44	55	52	36	71	55	44	40	60	60	53	50
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	19.3	22.3	25.3	27.1	30.6	33.8	34.2	35.6	33.1	29.9	24.2	20.7
		MCWB	15.1	16.8	16.9	18.1	21.4	22.7	23.3	24.1	21.0	19.7	15.8	17.2
	2%	DB	17.0	19.5	23.1	25.5	28.9	32.1	32.6	33.5	31.4	27.3	21.0	19.1
		MCWB	14.0	15.9	16.1	17.6	21.0	22.9	23.6	23.3	21.4	19.1	14.9	16.5
	5%	DB	15.0	17.6	21.2	24.3	27.6	31.0	31.4	31.8	30.0	25.4	19.3	17.2
		MCWB	12.0	13.9	14.8	17.2	20.7	22.8	23.7	23.1	21.5	17.9	14.7	14.9
	10%	DB	13.1	15.5	19.1	22.7	26.4	29.8	30.2	30.3	28.3	23.4	17.7	15.4
		MCWB	10.4	12.4	13.8	16.4	19.9	22.9	23.5	22.9	21.4	17.0	13.7	13.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	17.0	18.0	18.1	20.2	22.9	25.2	25.6	25.5	24.0	22.3	19.6	19.2
		MCDB	18.2	20.8	21.8	24.6	27.7	30.0	30.3	30.8	28.4	27.8	20.1	20.0
	2%	WB	14.8	16.9	17.2	19.3	22.1	24.5	25.0	24.8	23.4	21.2	17.8	17.5
		MCDB	15.8	19.3	21.1	23.4	27.0	29.7	29.9	29.9	28.2	25.3	19.3	19.0
	5%	WB	12.7	15.6	16.2	18.4	21.5	23.9	24.5	24.2	22.8	20.0	16.0	15.4
		MCDB	13.9	17.7	19.4	22.4	26.2	29.2	29.5	28.8	27.6	23.4	17.6	16.8
	10%	WB	10.3	13.5	15.1	17.6	20.9	23.3	24.0	23.7	22.1	18.5	14.1	13.5
		MCDB	12.0	15.1	18.2	21.5	25.3	28.5	28.9	28.0	26.3	21.6	16.7	15.0
Mean Daily Temperature Range	5% DB	MDBR	9.8	10.1	10.9	11.6	11.2	11.4	10.2	10.6	11.2	11.9	11.9	9.3
		MCDBR	11.0	11.4	12.4	12.7	12.6	13.3	12.0	12.8	13.5	13.5	13.1	9.9
		MCWBR	8.9	7.8	7.0	6.7	6.0	5.4	4.6	4.8	5.2	6.5	8.2	8.0
	5% WB	MCDBR	8.5	9.0	10.1	10.6	10.4	11.2	10.5	10.1	10.2	10.8	9.0	9.2
MCWBR		8.1	7.2	6.3	6.2	5.5	5.5	4.5	4.5	4.4	6.2	7.4	8.3	
Clear-Sky Solar Irradiance	taub	0.311	0.314	0.353	0.383	0.437	0.470	0.519	0.517	0.418	0.367	0.333	0.312	
	taud	2.519	2.506	2.419	2.318	2.210	2.154	2.035	2.028	2.295	2.428	2.466	2.531	
	Ebn at Noon	903	938	921	903	854	822	779	773	842	864	865	879	
	Edh at Noon	83	93	110	128	144	152	170	169	123	100	86	78	
All-Sky Solar Radiation	RadAvg	2.52	3.04	4.18	5.42	5.93	6.19	5.91	5.49	4.75	3.92	2.86	2.15	
	RadStd	0.18	0.39	0.39	0.36	0.47	0.48	0.38	0.30	0.43	0.50	0.32	0.21	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A
Regional (40 neighbors)	+0.49	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+153	+113

Nomenclature: See separate page